MANASH MISHRA

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience building production LLM and agentic systems. Built multi-agent platforms from scratch on MCP and LangGraph, with human-in-the-loop controls, evals, tracing (Langfuse) and guardrails, deployed on Docker and Kubernetes. Experienced in re-engineering manual business workflows into agent-driven processes, RAG and Text-to-SQL systems, and FinTech voice AI (ASR, TTS, LLM fine-tuning).

EXPERIENCE

AI Engineer, Dunnhumby, Gurgaon May 2025 – Present

- **Multi-Agent Measurement Platform:** Turned a manual 10-step campaign analysis workflow into an agent-driven process. Built an MCP-based agent harness from scratch that coordinates specialist agents for data extraction, SQL generation and impact analysis, with human-in-the-loop approval and Langfuse tracing.
- Deployed on Docker and Kubernetes for a Fortune 500 LATAM retailer and other global clients. Runs 1000+ campaign analyses a month, cut turnaround by 80% and reduced token spend by 40% through cost-aware model routing.
- **Text-to-SQL Platform:** Built a self-healing LLM query engine using FAISS schema retrieval (RAG), LangGraph and SQL guardrails. It serves 1000+ natural-language queries a day at 89% execution accuracy on a 500-question evaluation set.

Machine Learning Engineer, Paisabazaar, Gurgaon October 2023 – May 2025

- **Voice AI:** Architected a production voice bot for credit card sales and support on Azure, handling 10K+ calls a day with sub-second latency via a streaming ASR, LLM and TTS pipeline over WebSockets.
- Fine-tuned open-source Hindi TTS models for accurate financial-term pronunciation, serving 1000+ requests a day at a MOS of 4.3/5.
- **Compliance & Risk Analytics:** Built an automated call-quality pipeline for 100K+ calls using Kafka, fine-tuned Whisper ASR (40% relative WER reduction on Hindi audio) and LLM-based evaluation to flag do-not-call violations, fraud and abuse.
- Fine-tuned LLaMA with QLoRA on 10K+ labelled transcripts for fraud and do-not-call detection, reaching 92% F1 and beating prompt-based evaluation by 15%; deployed on AWS.

Research Associate, Indian Institute of Technology (IIT) BHU *July 2022 – September 2023*

Project sponsored by Bhashini, Govt. of India

- Built Hindi and Sanskrit semantic representation (USR) datasets and designed graph linearization algorithms with linguists, powering an interlingua-based MT system for the Government of India's National Translation Mission.
- Fine-tuned mT5 and BART for graph-to-sentence generation, and mT5 for Hindi POS tagging and complex predicate prediction in low-resource Indian languages (model link).

EDUCATION

Birla Institute Of Technology, Mesra

M.Tech Computer Science CGPA: 8.7

Sinhgad College of Engineering, Pune University

B.Tech Computer Science Percentage: 67

TECHNICAL SKILLS

Agentic AI: Multi-Agent Systems, Model Context Protocol (MCP), LangGraph, LangChain, LlamaIndex, Google ADK, Claude Code, Human-in-the-Loop Workflows

LLMs & RAG: FAISS, sentence-transformers, ColBERT, DSPy, GEPA

Evals & Observability: Langfuse, LLM Evaluation, Guardrails, Model Routing, Weights & Biases

Fine-Tuning: PyTorch, Hugging Face, PEFT (LoRA / QLoRA), Unsloth, Accelerate, Quantization

Inference & APIs: vLLM, Ollama, FastAPI, REST APIs, WebSockets

Speech AI: Whisper (ASR fine-tuning), Text-to-Speech, Real-time Audio Streaming, WER / MOS Evaluation